

```

clc;

clear;

close all;

%% --- Input Parameters ---

fp = 1000;    % Passband cutoff (Hz)

fs = 4000;    % Sampling frequency (Hz)

N = 50;      % Filter order (even number preferred)

wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- FIR Low Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'low', hamming(N+1));

disp('--- Hamming Window Coefficients ---');

disp(b_hamm);

%% --- Plot Frequency Responses ---

figure;

freqz(b_hamm,1);

title('Low-pass FIR using Hamming Window');

```

Output:

- Hamming Window Coefficients ---

0.0010

0

-0.0013

0

0.0021

0

-0.0034

0

0.0055

0

-0.0084

0

0.0125

0

-0.0181

0

0.0260

0

-0.0379

0

0.0580

0

-0.1026

0

0.3168

0.4995

0.3168

0

-0.1026

0

0.0580

0

-0.0379

0

0.0260

0

-0.0181

0

0.0125

0

-0.0084

0

0.0055

0

-0.0034

0

0.0021

0

-0.0013

0

0.0010

